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Measurement of the rotational velocity in Hanbit mirror device by using the Gundestrup-emissive-triple probe system

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A new probe set, called the Gundestrup-emissive-triple (GET) probe, is developed to measure the rotational velocity, plasma potential, electron temperature, and plasma density profiles simultaneously in Hanbit mirror device. The GET probe system is composed of the Gunderstrup, an emissive, and a triple probe installed in one boron nitride probe holder, which is attached on the fast scanning injection system. Rotational and parallel flow velocities are deduced from the measurement of Gundestrup probe by using fluid and kinetic theories for unmagnetized and magnetized plasmas. The $E\times B$ drift velocity is also deduced from the measurement of plasma potential by an emissive probe. Electron temperature and plasma density variations are measured by a triple probe. Applicability of various theories will be addressed. © *2004 American Institute of Physics.* [DOI: 10.1063/1.1789619]

I. INTRODUCTION

Plasma parameters, such as temperature, density and potential profiles are very important in understanding the transport properties and plasma dynamics in magnetic fusion devices and simulators, among which one is the Hanbit mirror devices,[1] and the plasma poloidal velocity and change of radial electric field are considered as essential parameters in plasma confinement. To measure the fundamental plasma parameters in the Hanbit device, a fast injection probe system[2] was installed in the central cell of Hanbit and the plasma parameters were measured by using a new probe set composed of emissive,[3] Gundestrup,[4] and triple probes loaded in the fast injection probe system. The Gundestrup probe is composed of several sets of Mach probes, which are located at various angles with respect to the direction of magnetic field. The Gundestrup-emissive-triple (GET) probe was designed to measure the plasma density, temperature, potential, and angular distribution of flow velocities along the magnetic field simultaneously.

A Mach probe is common method to measure plasma flow, which is composed of two Langmuir probes, shielded from each other, collecting ion saturation current from the upstream and the downstream directions. Although there have been many analyses of Mach probe theory in magnetized plasma,[5–7] there is no consistent theory for unmagnetized plasma.[8] However, the Mach probe data of poloidal direction in the central cell of Hanbit device had to be analyzed by the theory of unmagnetized plasma because the ion Lamor radius is larger than the probe radius and the flow direction is perpendicular to the B field.

To calculate the Match number, various unmagnetized Mach probe models, such as PIC,[9] and kinetic models,[10] are tried. To measure the profile of plasma potential in Hanbit, a strong emission method was used and the $E\times B$ drift velocity is deduced from potential variation.

II. EXPERIMENTAL SETUP

The main purpose of the design of the GET probe was to find an arrangement of Langmuir probes that could measure various plasma parameters simultaneously, and be mounted on a fast injection probe system. The GET probe head consists of an insulator tube and 12 different probe tips as shown in Fig. 1.

First, the Gundestrup probe is composed of eight probes along the periphery of the boron—nitride probe holder, which consists of four Mach probes. The Mach probe is often used to provide information about plasma flow velocities and typically consists of two identical collectors separated by an insulator. The two collectors of the Mach probe are made of tungsten wire 1 mm in diameter. Probe tips are 2.7 mm long and they are 45° apart along the poloidal direction. All of the eight collectors are negatively biased enough to draw the ion saturation currents and positioned to collect the directional charged particles. The ion saturation current I_{up} and I_{dn} for a Mach probe, are time averaged to reduce the fluctuation, where I_{up} and I_{dn} refer to upstream current and downstream current, respectively.

The GET probe head also has one emissive probe for measuring the plasma potential. The emissive probe is made of thoriated tungsten 0.24 mm in diameter, and is operated with a strong emission method which is much less sensitive to probe gemoetry and magnetic field direction and provides the plasma potential information. An additional probe is the triple probe measuring the plasma density and electron temperature directly. The tips of all three probes are positioned on the same flux surfaces. The parallel separation of each

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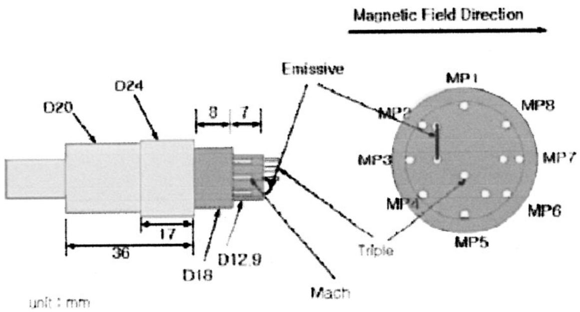

FIG. 1. GET probe configuration (triple probes: $d=1$ mm, $L=2.7$ mm, Mach probes: $d=1$ mm, $L=4.35$ mm, emissive probe: $d=0.24$ mm, $L=4$ mm).

probe is $\Delta z=3.2$ mm. Due to the size of the probe holder ($\sim$1.3 cm), plasma parameters of the wake region of the probe holder may be affected. Yet, if we take data when the probe is moving toward the center, the data of triple (density and temperature) and emissive (potential) probes located at the tip of the holder are not affected by the probe holder. However, those of Mach probes are affected by the size of the probe holder, which is intended for the asymmetry of current ratio (ram/wake) for the deduction of the Mach number.

As a regular operation mode of the Hanbit mirror device, a simple wave form of rf power is applied. The rf power is ramped up to 100 kW from 0 to 20 ms and remains at this level to 300 ms, then is ramped down to zero at 350 ms. Although the rf frequency (3.5 MHz) is very close to ion cyclotron resonance frequency (for $B=2.4-2.6$ T) the plasma production mechanism are as follows: rf energy goes to the electron first through the electron Landau damping to ionize hydrogen atoms then increases electron temperature. Because the propagating wave is a fast wave and it does not have an electric field polarized in an ion sense near the central plasma region, centralized ion heating cannot be expected. For $\omega > \omega_{ci}$, a slow wave cannot be propagated to cause any heating effect and ion heating happens through the weak kinetic effect on the dispersion relation and the electron drag.

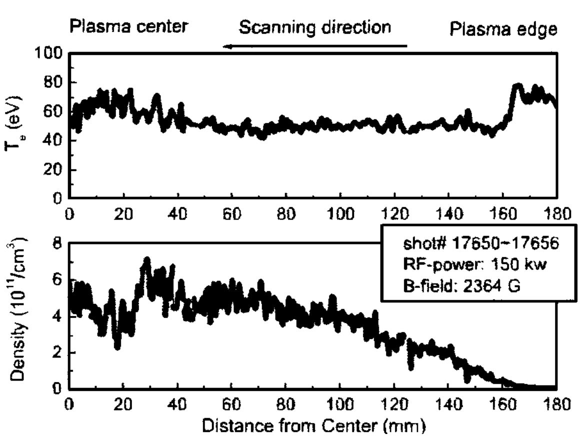

FIG. 2. The results of the electron temperature and density measured by triple probe.

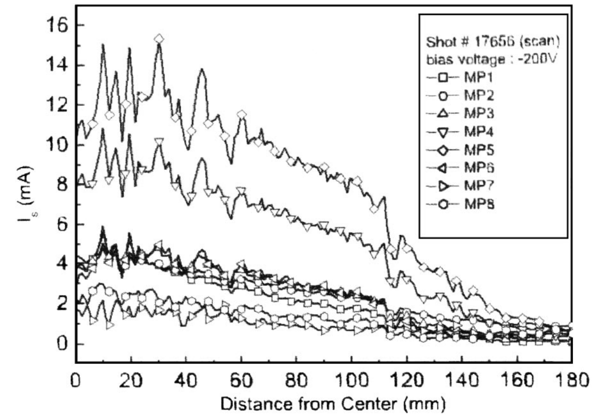

FIG. 3. Radial profiles of ion saturation currents.

III. RESULTS

Figure 2 shows the radial profiles of electron temperature and plasma density profile calculated from the ion saturation current near 100 ms during the discharge for rf power of 150 kW and magnatic field of 0.24 T in the central cell of the Hanbit device. The line intensity and the net rf input power showed good reproducibility near 100 ms. Toward the center of the chamber, the electron temperature decreases while electron density increases after a sudden drop around $r=30$ mm, which is not clear with the current data. More relevant data are required for further analysis.

Figure 3 shows the ion saturation currents collected by the eight probe tips of the Gundestrup probe.

Figure 4 shows the ratios of four Mach probes, i.e., MP5/MP1, MP6/MP2, MP3/MP7, and MP4/MP8. From the data of MP3/MP7, one can observe that there is almost uniform flow along the magnetic field, although this axial flow slightly decreases near the edge. By using fluid and kinetic models for magnetized plasma,[6,7] axial Mach numbers for MP7/MP3 are 0.17–0.30, as shown in Fig. 5. From the data of MP5/MP1 and MP4/MP8 in Fig. 4, we can see that the rotational velocity at the center is not always zero, but rather there are flows toward 0° (upward) and 45° (right upward). However, data of MP6/MP2 show that zero flows toward the 315° direction. The Mach number deduced by the data of MP5/MP1 and MP6/MP2 applying the kinetic and

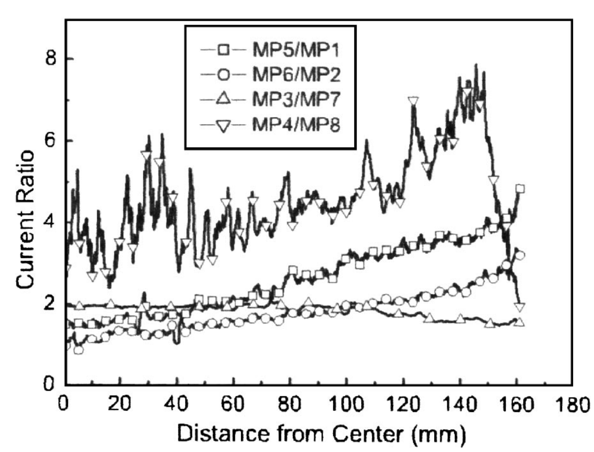

FIG. 4. Ion saturation current ratio of Gunderstrup probe.

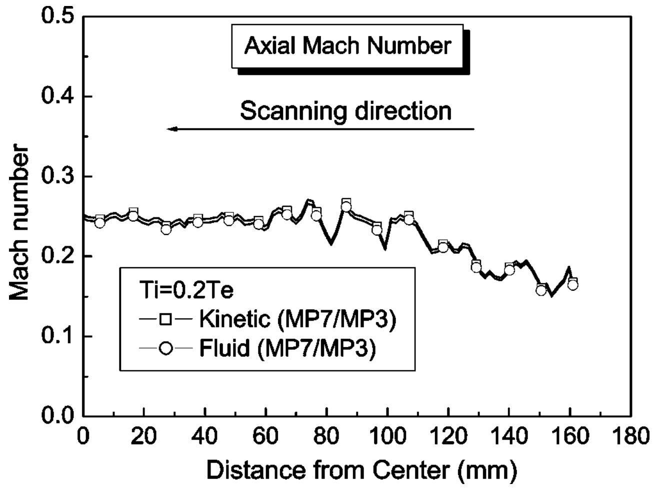

FIG. 5. Radial profile of axial Mach number.

PIC results,[9,10] for unmagnetized plasmas, are shown in Fig. 6. Although the two models were treated with different geometry of probes (plane and sphere), with different analyses, the results are similar ($R=\exp[kM]$, $M=V_d/\sqrt{T_e/m_i}$, $k=1.2$ for kinetic and 1.34 for PIC) and this indicates the reliability of two models. Figure 7 shows the $E\times B$ drift velocities deduced from the plasma potential profiles which are measured by an emissive probe. These are much smaller than those by the Mach probes, which indicated an additional driving mechanism for strong rotation in the Hanbit device.

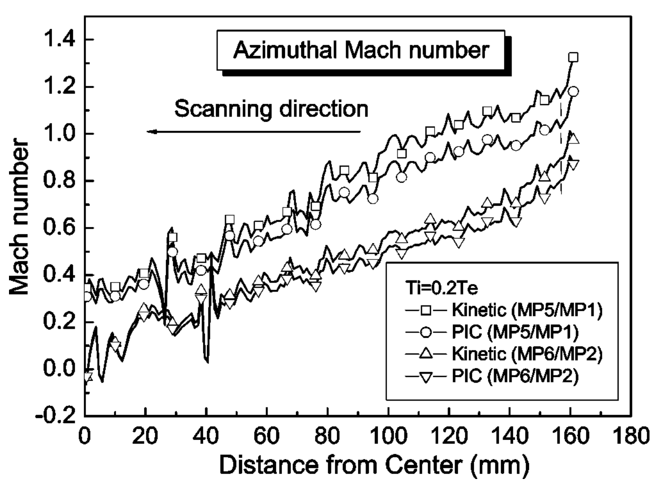

FIG. 6. Radial profiles of poloidal Mach number.

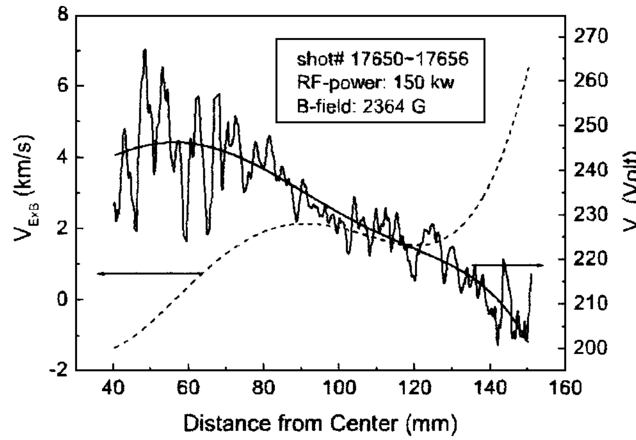

FIG. 7. $E\times B$ drift velocities deduced by potential measurement.

IV. DISCUSSION

For the first time, a versatile GET probe system has been developed to measure the parameters of the Hanbit plasma simultaneously. The radial plasma potential and poloidal velocity are analyzed under the same central cell magnetic field and rf power wave form. The poloidal Mach number has been measured in the central cell of the Hanbit device by using kinetic and PIC models for the unmagnetized plasmas with lower ion temperature, and they produce similar results confirming the reliability of two models, although they adopted different geometries (plane and sphere). A smaller value of $E\times B$drift requires an additional mechanism for rotation. Mach probe results are recommended to be confirmed by another diagnostic such as laser-induced fluorescence or optical emission spectroscopy.

ACKNOWLEDGMENT

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